Supply Radar — Results & Implications

Use Case 6 · Supplier & Sourcing Intelligence for Vaighai · summary for the team

The solution

Supply Radar combines the MIR field estimates, our ERP purchase records, and regional rainfall into one supplier view. It flags which mills are likely to drop next quarter, forecasts how much each will supply, ranks where we should buy more, and tracks how dependent we are on a few mills. It runs as a live dashboard with a weekly plain-English brief, fully local today and ready to move to Azure.

Key results (next quarter, FY2027 FQ2)

- **510 suppliers** scored. Total forecast supply next quarter **~50,800 MT** (~39,800 MT after allowing for drop-risk).
- **27 mills at critical risk** of a sharp drop, 92 at moderate risk — most stopped supplying last quarter and we already take 0% of them.
- Biggest volume at risk: **Vishvajith Fiber, Sri Krishna Coir Impexs, Kadeswaran Pith Block, Sabarigiri Fibres** — worth a call/visit before they are lost.
- Top sourcing opportunities (big, fast-growing mills we barely buy from): **Vetrivel Pith Block (2,819 MT), Balu Coir (2,501), Selvanayagiamman (2,320), Palanimurugan (1,720)** — concentrated in **Kangeyam and Salem**.
- Supplier concentration has **halved**: top-5 dependence fell from ~91% (FY2022) to **51% (FY2026)** — we are far less exposed to a few mills.
- But our own purchase volume has **fallen ~78% from its FY2022 peak** (~324k to ~72k MT/yr) — the key strategic question to investigate.
- Weather: a **dry fortnight ahead** across all regions — production and drying should run freely near-term.

What to do

- **Grow:** pursue the Kangeyam & Salem opportunity list — that is where the largest untapped, growing volume sits.
- **Protect:** field-check the big at-risk mills this week to confirm status and lock volume.
- **Investigate:** why our sourcing volume dropped ~78% since FY2022 — biggest lever on the business.
- Use the risk flags and opportunity ranking to drive weekly buying; treat exact forecast tonnage as guidance (small mills are volatile).

How reliable

Decline-risk model cross-validated AUC 0.71 (precision 0.65, recall 0.44). Volume forecasts are directional, not exact — supplier-level quarterly demand swings a lot. The strongest, most actionable outputs are the risk bands and the sourcing shortlist.

Deliverables: live dashboard (React) · detailed per-supplier forecast (Excel) · weekly AI sourcing brief · full speaker notes.